

Algebra Essentials



Evaluating expressions

- 1 If $x = 4$, evaluate $3x + 7$.
 - 2 If $a = 5$ and $b = 2$, evaluate $a^2 - 3b$.
 - 3 If $x = -3$, evaluate $2x^2 + 5x$.
 - 4 If $p = 6$ and $q = -2$, evaluate $\frac{p - q}{q + 6}$.
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Solving linear equations

- 5 Solve $2x + 5 = 17$.
 - 6 Solve $5x - 8 = 27$.
 - 7 Solve $3(x - 4) = 21$.
 - 8 Solve $\frac{x}{4} + 3 = 10$.
 - 9 Solve $7x + 3 = 4x + 18$.
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Expanding and simplifying

- 10 Expand $(x + 3)(x + 5)$.
 - 11 Expand $(x - 4)(x + 6)$.
 - 12 Expand $(2x + 1)(x - 3)$.
 - 13 Simplify $3(2x - 5) - 2(x - 4)$.
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Factoring

- 14 Factor $x^2 + 9x + 20$.
- 15 Factor $x^2 - 2x - 15$.

16 Factor $x^2 - 49$.

17 Factor $6x + 15$.

Solving quadratic equations by factoring

18 Solve $x^2 + 5x + 6 = 0$.

19 Solve $x^2 - 7x + 12 = 0$.

20 Solve $x^2 - 16 = 0$.

21 Solve $x^2 + 2x - 15 = 0$.

Systems of two equations

22 Solve the system: $x + y = 10$, $x - y = 4$.

23 Solve the system: $2x + y = 11$, $x - y = 1$.

24 Solve the system: $x = 3y - 1$, $2x + y = 19$.

25 Solve the system: $4x - y = 10$, $2x + y = 8$.

26 Solve the system: $5x + 3y = 21$, $2x - 3y = 0$.

Laws of exponents

27 Simplify $a^5 \times a^3$.

28 Simplify $\frac{x^9}{x^4}$.

29 Simplify $(y^3)^4$.

30 Simplify $2x^3 \times 5x^2$.

Inequalities

- 31 Solve $3x - 4 > 11$.
- 32 Solve $-2x + 6 \leq 20$.
- 33 Solve $5x + 1 \geq 3x + 9$.
- 34 Solve $4(x - 2) < 2x + 6$.
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Substitution into formulas

- 35 The formula for simple interest is $I = Prt$. Find I when $P = 2,000$, $r = 0.04$, $t = 3$.
- 36 The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.
- 37 If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.
- 38 The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.
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Translating words into equations

- 39 Twice a number, plus 7, equals 25. Find the number.
- 40 A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.
- 41 The sum of three consecutive integers is 84. Find the integers.
- 42 Five more than three times a number is the same as 10 less than five times the number. Find the number.
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Word problems

- 43 A rectangle has length $(x + 3)$ cm and width x cm. This question has two parts.
- a If the area is 40 cm^2 , write an equation in x .
- b Solve for x .
- 44 A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.
- 45 Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

- 46 The sum of two numbers is 45, and their difference is 11. Find the two numbers.
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MCQ — Evaluating expressions

- 47 If $x = 6$, evaluate $4x - 9$.
- a. 33
 - b. 24
 - c. -15
 - d. 15
- 48 If $a = -2$ and $b = 3$, evaluate $a^2 + 2ab$.
- a. -16
 - b. 8
 - c. -4
 - d. -8
- 49 If $x = 5$ and $y = -1$, evaluate $(x + y)^2 - xy$.
- a. 16
 - b. 11
 - c. 21
 - d. -5

50 If $p = 3$, evaluate $2p^3 - 5p$.

a. 49

b. 39

c. 33

d. 69

MCQ — Solving linear equations

51 Solve $\frac{x}{3} + 4 = 9$.

a. $x = 27$

b. $x = 15$

c. $x = 5$

d. $x = 39$

52 Solve $4(x + 2) = 3x + 15$.

a. $x = 1$

b. $x = 23$

c. $x = -7$

d. $x = 7$

53 Solve $0.5x - 3 = 7$.

a. $x = 20$

b. $x = 8$

c. $x = -20$

d. $x = 14$

54 Solve $\frac{2x-1}{3} = 5$.

a. $x = 4$

b. $x = 7$

c. $x = 16$

d. $x = 8$

55 Solve $6 - 2x = 3x - 9$.

a. $x = -3$

b. $x = 3$

c. $x = 15$

d. $x = 1$

MCQ — Expanding and simplifying

56 Expand $(x + 7)^2$.

a. $x^2 + 7x + 49$

b. $x^2 + 14x + 14$

c. $x^2 + 14x + 49$

d. $x^2 + 49$

57 Expand $(x - 5)^2$.

a. $x^2 + 10x + 25$

b. $x^2 - 5x + 25$

c. $x^2 - 10x + 25$

d. $x^2 - 25$

58 Expand $(3x - 2)(3x + 2)$.

a. $9x^2 + 4$

b. $9x^2 - 12x - 4$

c. $9x^2 - 4$

d. $6x^2 - 4$

59 Simplify $2(x + 3) - 3(x - 1)$.

a. $5x + 9$

b. $-x + 9$

c. $x + 9$

d. $-x + 3$

MCQ — Factoring

60 Factor $3x^2 + 12x$.

a. $3(x^2 + 4x)$

b. $x(3x + 12)$

c. $3x(x + 12)$

d. $3x(x + 4)$

61 Factor $x^2 - 11x + 30$.

a. $(x - 5)(x - 6)$

b. $(x - 2)(x - 15)$

c. $(x + 5)(x + 6)$

d. $(x - 3)(x - 10)$

62 Factor $x^2 + 3x - 18$.

a. $(x + 2)(x - 9)$

b. $(x - 6)(x + 3)$

c. $(x + 9)(x - 2)$

d. $(x + 6)(x - 3)$

63 Factor $4x^2 - 25$.

a. $(2x - 25)(2x + 1)$

b. $(2x - 5)(2x + 5)$

c. $(2x - 5)^2$

d. $(4x - 5)(x + 5)$

MCQ — Solving quadratic equations

64 Solve $x^2 - 9x + 18 = 0$.

a. $x = -3$ or $x = -6$

b. $x = 1$ or $x = 18$

c. $x = 2$ or $x = 9$

d. $x = 3$ or $x = 6$

65 Solve $x^2 + 2x - 24 = 0$.

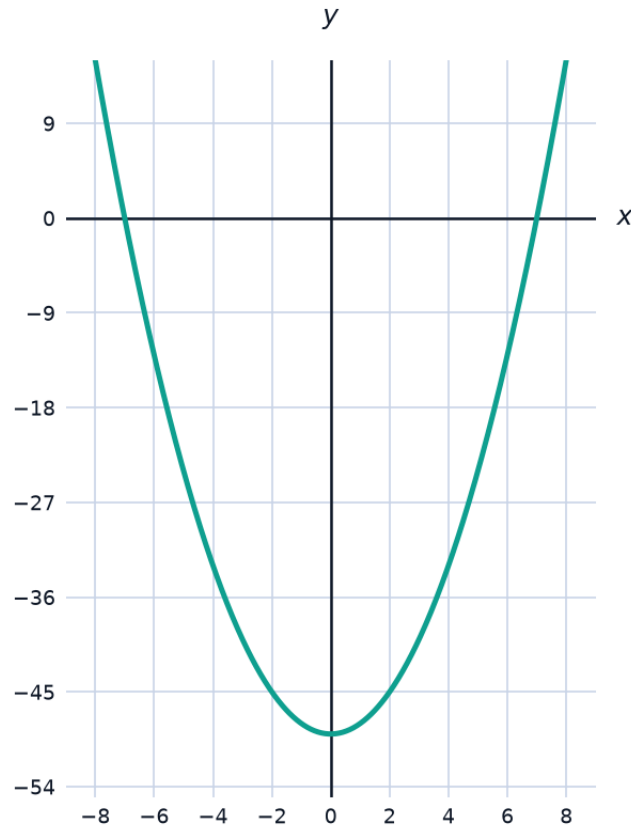
a. $x = 6$ or $x = -4$

b. $x = -2$ or $x = 12$

c. $x = -6$ or $x = 4$

d. $x = -3$ or $x = 8$

- 66 The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x -axis)?



- a. $x = 7$ only
 - b. $x = 49$
 - c. $x = 7$ or $x = -7$
 - d. $x = -7$ only
- 67 Solve $2x^2 - 8x = 0$.
- a. $x = 8$
 - b. $x = 2$ or $x = 4$
 - c. $x = 0$ or $x = 4$
 - d. $x = 0$ or $x = -4$

MCQ — Systems of two equations

68 Solve the system: $x + 2y = 8$, $x - y = 2$.

a. $x = 2$, $y = 3$

b. $x = 6$, $y = 1$

c. $x = 4$, $y = 2$

d. $x = 8$, $y = 0$

69 Solve the system: $3x - y = 7$, $x + y = 5$.

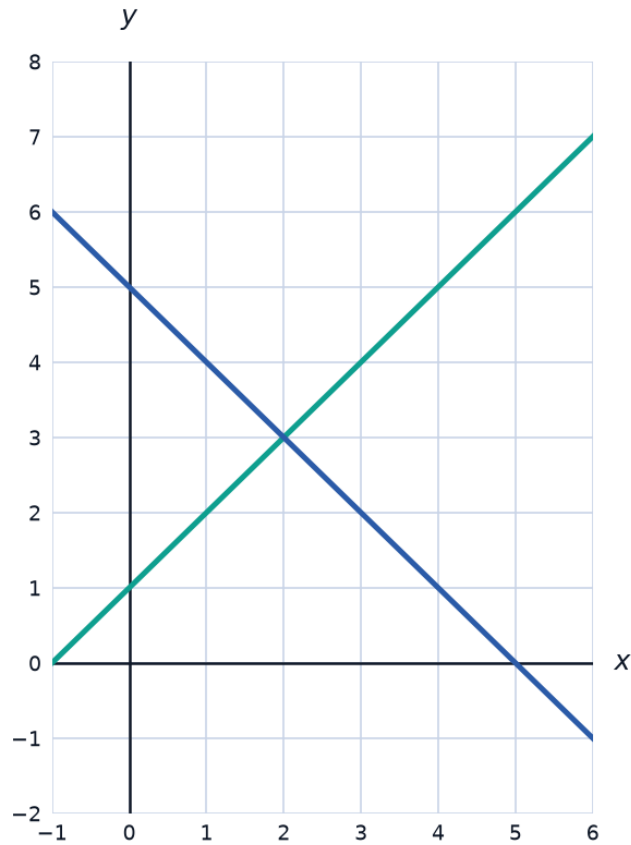
a. $x = 2$, $y = 3$

b. $x = 1$, $y = 4$

c. $x = 3$, $y = 2$

d. $x = 4$, $y = 1$

70 The graph shows two lines. What is their point of intersection?



a. (1, 4)

b. (0, 5)

c. (2, 3)

d. (3, 2)

71 Solve the system: $y = 2x - 1$, $y = x + 3$.

a. $x = -4$, $y = -5$

b. $x = 2$, $y = 3$

c. $x = 7$, $y = 4$

d. $x = 4$, $y = 7$

MCQ — Laws of exponents

72 Simplify $b^7 \div b^2$.

a. $b^{3.5}$

b. b^{14}

c. b^9

d. b^5

73 Simplify $(2x^3)^2$.

a. $4x^6$

b. $2x^9$

c. $4x^5$

d. $2x^6$

74 Simplify $x^0 \cdot x^4$ ($x \neq 0$).

a. 1

b. x^5

c. x^0

d. x^4

75 Simplify m^{-2} .

a. $-\frac{1}{m^2}$

b. $\frac{1}{m^2}$

c. m^2

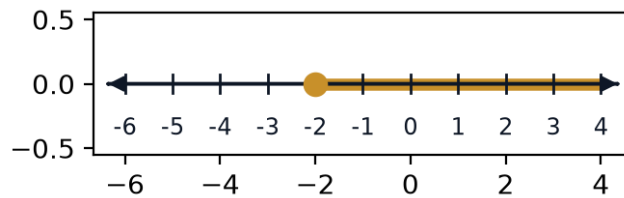
d. $-m^2$

MCQ — Inequalities

- 76 The number line shows the solution set of an inequality. Which inequality matches?



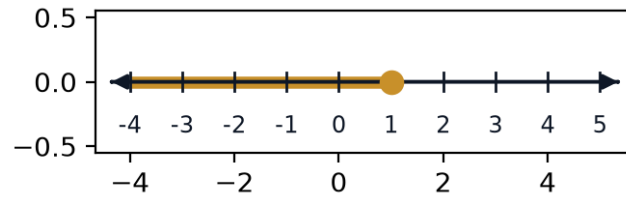
- a. $x \geq 4$
 - b. $x \leq 4$
 - c. $x > 4$
 - d. $x < 4$
- 77 The number line shows the solution set of an inequality. Which inequality matches?



- a. $x \leq -2$
 - b. $x < -2$
 - c. $x \geq -2$
 - d. $x > -2$
- 78 Solve $-3x + 7 \leq 22$.

- a. $x \geq -5$
- b. $x \leq -5$
- c. $x \leq 5$
- d. $x \geq 5$

79 The number line shows the solution set of an inequality. Which inequality matches?



a. $x > 1$

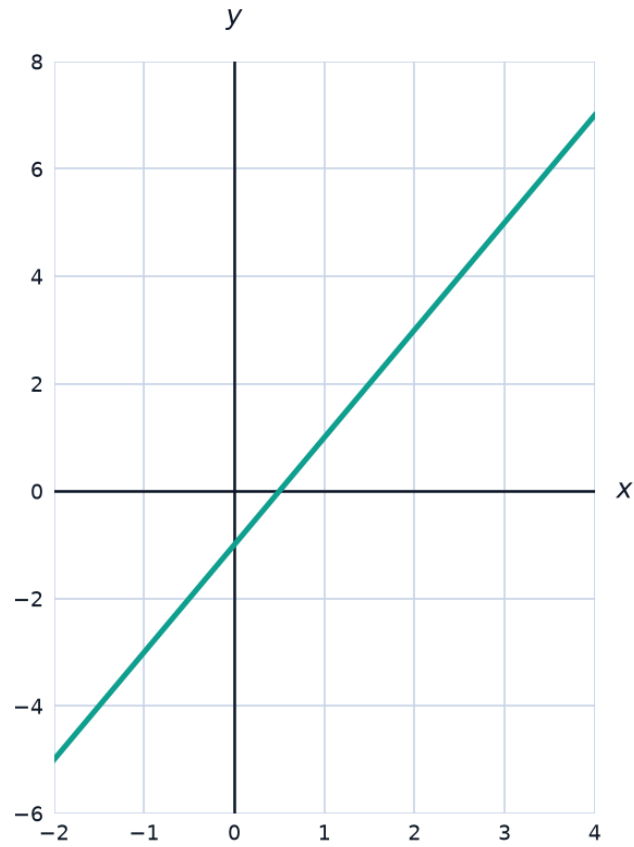
b. $x \geq 1$

c. $x \leq 1$

d. $x < 1$

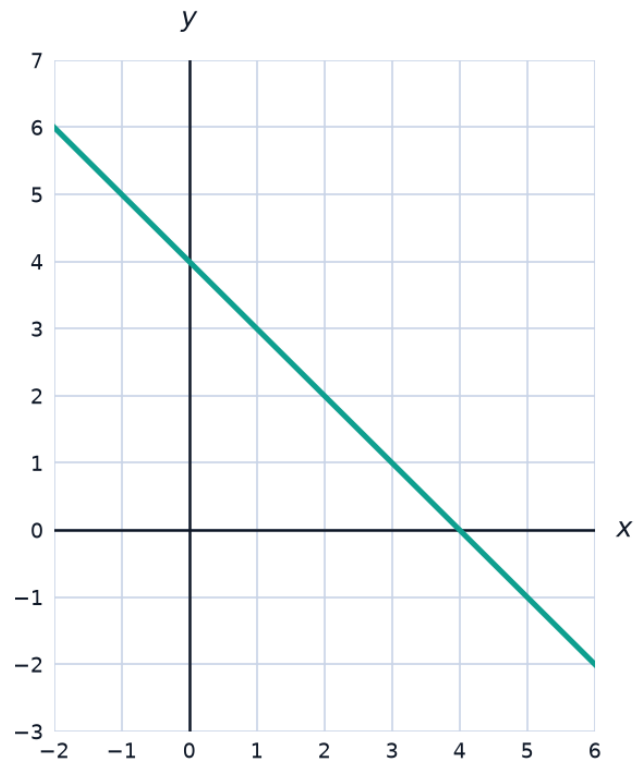
MCQ — Reading function graphs

80 The graph shows a line. What is its slope?



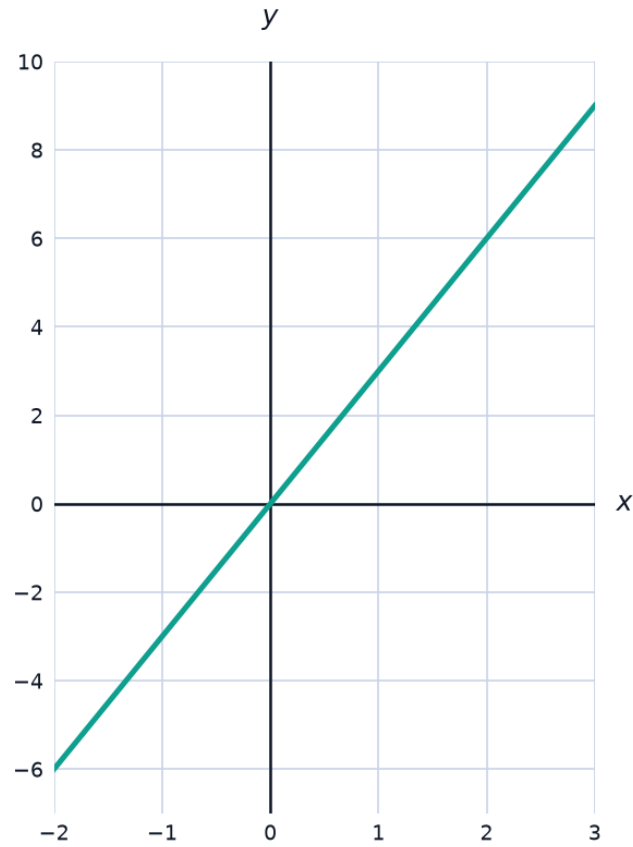
- a. $\frac{1}{2}$
- b. 2
- c. -2
- d. -1

81 What is the y-intercept of the line shown?



- a. 4
- b. 0
- c. -4
- d. 1

82 What is the equation of the line shown?



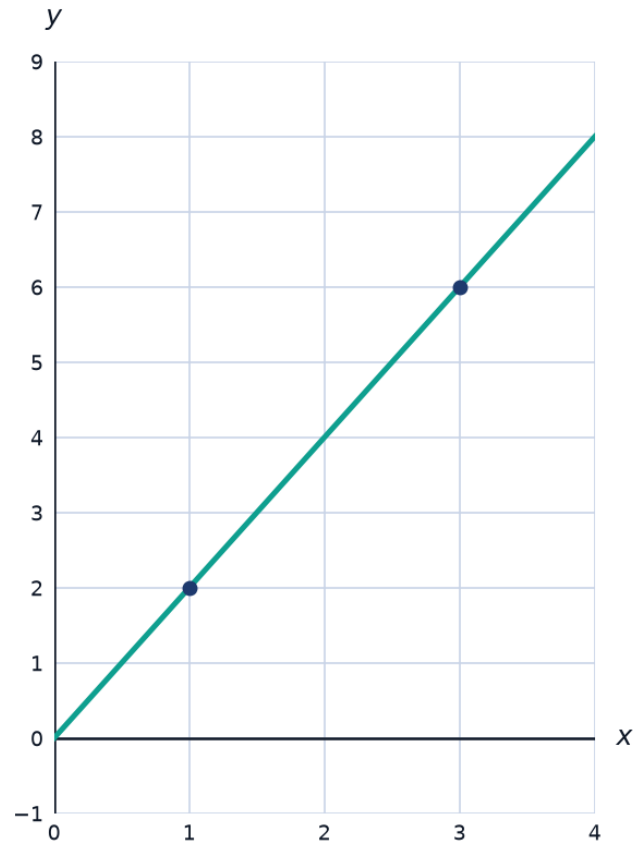
a. $y = 3x + 1$

b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$

- 83 Find the slope of the line through the two marked points.



- a. 2
- b. $\frac{1}{2}$
- c. -2
- d. 4

MCQ — Substitution into formulas

- 84 Given $A = \frac{1}{2}bh$, find A when $b = 10$ and $h = 6$.

- a. 16
- b. 8
- c. 60
- d. 30

85 Given $KE = \frac{1}{2}mv^2$, find KE when $m = 4$ and $v = 3$.

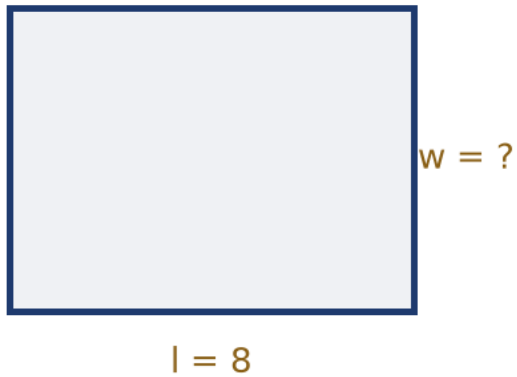
a. 18

b. 12

c. 24

d. 36

86 The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w .



a. 8

b. 6

c. 384

d. 40

87 Given $F = ma$, find F when $m = 7$ and $a = 5$.

a. 35

b. 75

c. 12

d. 2.4

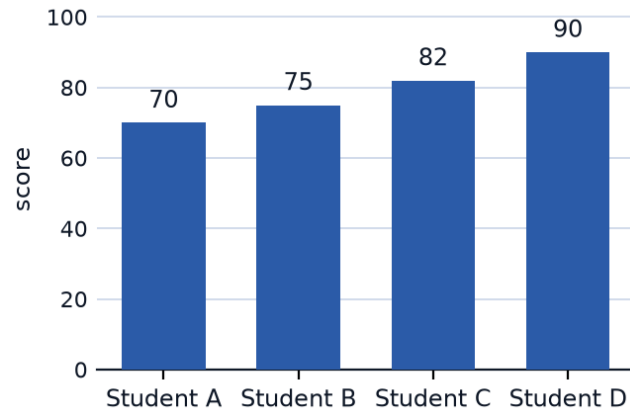
MCQ — Translating words into equations

- 88** Three times a number, minus 5, equals twice the number plus 7. Find the number.
- a. 19
 - b. 2
 - c. -12
 - d. 12
- 89** The sum of two consecutive even integers is 54. Find the larger integer.
- a. 28
 - b. 27
 - c. 29
 - d. 26
- 90** Five less than three times a number is 25. Find the number.
- a. 10
 - b. 20
 - c. 5
 - d. 15
- 91** The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.
- a. 24
 - b. 12
 - c. 15
 - d. 9

MCQ — Word problems

- 92 A number increased by 20% equals 72. Find the number.
- a. 90
 - b. 52
 - c. 60
 - d. 86.4
- 93 Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.
- a. 20
 - b. 10
 - c. 15
 - d. 25
- 94 A positive number plus its square is 30. Find the number.
- a. 30
 - b. -6
 - c. 6
 - d. 5

- 95 The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?



- a. Student B
 - b. Student D
 - c. Student C
 - d. Student A
- 96 The product of two consecutive integers is 132. Find the larger integer.
- a. 13
 - b. 12
 - c. 10
 - d. 11

Multiple choice — easy

- 97 Solve $9x - 15 = 3$.
- a. 2
 - b. 4
 - c. -2
 - d. 0

98 Solve $9x + 12 = -24$.

a. -4

b. -2

c. 4

d. -6

99 Solve $8x + 13 = 117$.

a. -13

b. 13

c. 11

d. 15

100 Solve $8x - 20 = -140$.

a. -17

b. -15

c. 15

d. -13

101 Solve $9x - 12 = -147$.

a. -17

b. -15

c. 15

d. -13

102 Solve $8x + 10 = 114$.

- a. 13
- b. 11
- c. -13
- d. 15

103 Solve $9x + 15 = 150$.

- a. 15
- b. -15
- c. 17
- d. 13

104 Solve $9x + 1 = 28$.

- a. -3
- b. 3
- c. 5
- d. 1

105 Solve $2x = 6$.

- a. 3
- b. 5
- c. 1
- d. -3

106 Solve $7x + 4 = -3$.

- a. 1
- b. 3
- c. -3
- d. -1

107 Solve $5x - 12 = -72$.

- a. -10
- b. -14
- c. -12
- d. 12

108 Solve $3x + 12 = 21$.

- a. -3
- b. 5
- c. 3
- d. 1

109 Solve $5x + 18 = -27$.

- a. -9
- b. 9
- c. -7
- d. -11

110 Solve $2x = -14$.

- a. -7
- b. -5
- c. 7
- d. -9

111 Solve $5x - 10 = 20$.

- a. 8
- b. 4
- c. 6
- d. -6

Multiple choice — medium

112 Expand $(x + 4)(x - 2)$.

- a. $x^2 + 2x - 8$
- b. $x^2 - 2x - 8$
- c. $x^2 + 2x + 8$
- d. $x^2 - 8x + 2$

113 Expand $(x + 5)(x + 7)$.

- a. $x^2 - 12x + 35$
- b. $x^2 + 12x - 35$
- c. $x^2 + 12x + 35$
- d. $x^2 + 35x + 12$

114 Expand $(x + 3)(x + 7)$.

a. $x^2 + 21x + 10$

b. $x^2 - 10x + 21$

c. $x^2 + 10x - 21$

d. $x^2 + 10x + 21$

115 Expand $(x + 9)(x - 6)$.

a. $x^2 + 3x - 54$

b. $x^2 - 3x - 54$

c. $x^2 - 54x + 3$

d. $x^2 + 3x + 54$

116 Expand $(x + 1)(x - 1)$.

a. $x^2 + 2x - 1$

b. $x^2 - x$

c. $x^2 + 1$

d. $x^2 - 1$

117 Expand $(x + 2)(x + 2)$.

a. $x^2 - 4x + 4$

b. $x^2 + 6x + 4$

c. $x^2 + 4x - 4$

d. $x^2 + 4x + 4$

118 Expand $(x - 4)(x - 3)$.

a. $x^2 + 12x - 7$

b. $x^2 - 7x + 12$

c. $x^2 + 7x + 12$

d. $x^2 - 7x - 12$

119 Expand $(x - 2)(x - 1)$.

a. $x^2 + 2x - 3$

b. $x^2 + 3x + 2$

c. $x^2 - 3x + 2$

d. $x^2 - 3x - 2$

120 Expand $(x + 9)(x - 2)$.

a. $x^2 - 7x - 18$

b. $x^2 + 7x + 18$

c. $x^2 - 18x + 7$

d. $x^2 + 7x - 18$

121 Expand $(x - 5)(x + 9)$.

a. $x^2 + 4x - 45$

b. $x^2 - 45x + 4$

c. $x^2 + 4x + 45$

d. $x^2 - 4x - 45$

122 Expand $(x - 2)(x - 8)$.

a. $x^2 - 10x + 16$

b. $x^2 + 10x + 16$

c. $x^2 + 16x - 10$

d. $x^2 - 10x - 16$

123 Expand $(x - 4)(x + 7)$.

a. $x^2 - 28x + 3$

b. $x^2 + 3x + 28$

c. $x^2 - 3x - 28$

d. $x^2 + 3x - 28$

124 Expand $(x + 8)(x - 8)$.

a. $x^2 + 2x - 64$

b. $x^2 - 64$

c. $x^2 - 64x$

d. $x^2 + 64$

125 Expand $(x + 4)(x - 3)$.

a. $x^2 - 12x + 1$

b. $x^2 + x + 12$

c. $x^2 - x - 12$

d. $x^2 + x - 12$

126 Expand $(x - 1)(x - 1)$.

a. $x^2 - 2x + 1$

b. $x^2 - 2x - 1$

c. $x^2 + 2x + 1$

d. $x^2 + x - 2$

Multiple choice — hard

127 Solve $x^2 - 9x + 14 = 0$.

a. $-2, -7$

b. $2, 8$

c. $3, 7$

d. $2, 7$

128 Solve $x^2 - 14x + 48 = 0$.

a. $6, 9$

b. $-6, -8$

c. $6, 8$

d. $7, 8$

129 Solve $x^2 - 25 = 0$.

a. $5, -5$

b. $-5, 5$

c. $6, -5$

d. $5, -4$

130 Solve $x^2 - 12x + 35 = 0$.

- a. 8, 5
- b. -7, -5
- c. 7, 5
- d. 7, 6

131 Solve $x^2 + 10x + 16 = 0$.

- a. -2, -7
- b. 2, 8
- c. -2, -8
- d. -1, -8

132 Solve the system: $x - y = -1$, $x + y = -17$.

- a. -8, -9
- b. -9, -8
- c. -9, -6
- d. -7, -8

133 Solve the system: $3x - y = -31$, $x + y = -5$.

- a. 4, -9
- b. -9, 6
- c. -7, 4
- d. -9, 4

134 Solve the system: $x + y = -3$, $x - y = 9$.

- a. 3, -4
- b. -6, 3
- c. 3, -6
- d. 5, -6

135 Solve the system: $x + y = -8$, $x - y = -10$.

- a. 1, -9
- b. -9, 1
- c. -9, 3
- d. -7, 1

136 Solve the system: $x - y = 5$, $x + y = -11$.

- a. -3, -6
- b. -8, -3
- c. -1, -8
- d. -3, -8

137 Simplify $a^4 \times a^8$.

- a. a^{32}
- b. a^{11}
- c. a^{12}
- d. a^{13}

138 Simplify $\frac{x^9}{x^5}$.

a. x^5

b. x^4

c. x^3

d. x^{45}

139 Simplify $(b^4)^3$.

a. b^{12}

b. b^7

c. b^{14}

d. b^{10}

140 Simplify $a^4 \times a^5$.

a. a^8

b. a^{10}

c. a^{20}

d. a^9

141 Simplify $\frac{x^7}{x^2}$.

a. x^{14}

b. x^5

c. x^6

d. x^4